17. Median of two Sorted ArrMedian of two Sorted Arrays of Different Sizes

```
class Solution {
 public:
  double medianOf2(vector<int>& a, vector<int>& b) {
      int n1 = a.size(), n2 = b.size();
      vector<int> merged;
      merged.reserve(n1 + n2);  int i = 0, j = 0;
      while (i < n1 && j < n2) {
          if (a[i] <= b[j])
              merged.push_back(a[i++]);
          else
              merged.push_back(b[j++]);
      } while (i < n1) merged.push_back(a[i++]);
      while (j < n2) merged.push_back(b[j++]);

      int n = merged.size();
      if (n % 2 == 1)
          return merged[n / 2];
      else
          return (merged[n / 2 - 1] + merged[n / 2]) / 2.0;
  }
};
```